

National Institute of Technology, Kurukshetra
B.Tech (4th Sem)
(ITPC 200 Operating Systems) Information Technology/
(CSPC 200 Operating Systems) Mathematics and Computing/
(AIPC 202 Operating Systems) Artificial Intelligence and Machine learning

Duration: 3 Hours.

Max.Marks:50

Note: Attempt any five questions. Each question carries 10 marks each.

Q1 Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. How many page faults would occur for the following replacement algorithms, assuming demand paging with seven frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each. a) LRU replacement b) FIFO replacement c) Optimal replacement

Q2. Select any one classical synchronization problem (Producer-Consumer, Readers-Writers, Dining Philosophers, etc.). Design a synchronization algorithm using suitable primitives such as semaphores, monitors, or lock-free constructs. Prove that your solution satisfies mutual exclusion, progress, and bounded waiting, and analyze its susceptibility to deadlock and starvation.

Or

What is optimistic assumption made in deadlock detection algorithm? How could that assumption be violated?

Q3. Assume that amount of memory on a system is inversely proportional to the page fault rate. Each time page fault rate doubles, when memory is reduced by half. Currently the system has 128 MB of memory. The main memory access time is 1 μ sec and page fault service time is 1 msec. The overall effective access time is 112 μ sec. If the memory is decreased to 32 MB then what will be the overall effective access time?

Q4. Evaluate the suitability of Tree-Structured, Acyclic Graph, and General Graph directory organizations for contemporary operating systems. Compare these directory structures based on search efficiency, file sharing, storage requirements, security, fault tolerance, and scalability. Based on your analysis, recommend the most appropriate directory organization for a cloud-based file system and justify your recommendation.

Q5. Access Control Lists (ACLs) are widely used mechanisms for implementing protection in modern operating systems. Explain the design and operation of ACL-based protection systems. Critically analyse their scalability, security limitations, and suitability for distributed and cloud computing environments.

Q6 Analyse the interplay between long-term, medium-term, and short-term scheduling in achieving efficient process management and resource utilization. Why it is important for a scheduler to distinguish I/O bound programs from CPU bound Programs?

Q7. Although the Shortest Seek Time First (SSTF) algorithm minimizes immediate disk head movement and often provides better average seek performance, it does not guarantee optimal overall disk scheduling. Critically analyse this statement by discussing its advantages, limitations, starvation issues, and impact on fairness

Q8. Discuss the fundamental limitations and trade-offs involved in achieving complete transparency in distributed systems, considering issues such as scalability, performance overhead, consistency, fault tolerance, and security.